

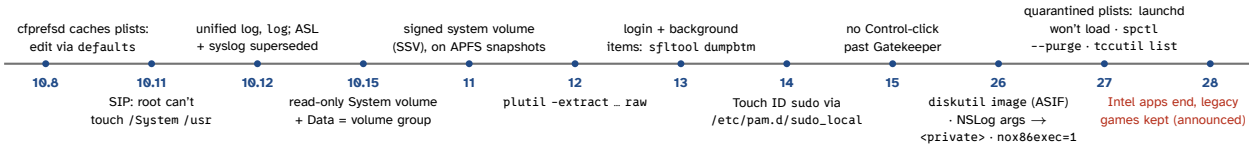
In one line: Most macOS admin commands are **thin clients of a daemon** that owns and caches the state — so change it *through the tool*, name the **domain** (prefs domain, launchd gui/<uid>, log subsystem), and expect **SIP**, **TCC** and the **sealed system volume** to refuse even root.

your shell — each tool asks the daemon (*italic*) that owns the state; the daemon, not the file on disk, is the truth

Preferences <i>cfprefsd</i> defaults · UserDefaults plutil = raw file!	Services <i>launchd (PID 1)</i> launchctl · sftool brew services	Logs <i>logd</i> log stream / show log collect / config	Spotlight <i>mds · mdworker</i> mdfind · mdls mdutil -s / -E	Apps & clipboard <i>lsd · pboard</i> open -a -b -n -W -R pbcopy · pbpaste	Power <i>powerd</i> pmset -g assertions caffeinate -dimsu
Storage <i>diskarbitrationd · backupd</i> diskutil · hdiutil tmutil · cp -c	Network <i>configd · mDNSResponder</i> scutil · networksetup dscacheutil -q host	Users & groups <i>opendirectoryd</i> dscl . -read /Users/x dseditgroup · id	System info <i>kernel · sysctl MIB</i> sysctl · sw_vers · ioreg system_profiler	Security <i>syspolicyd · TCC</i> spctl -a · xattr csrutil · tcutil	Process inspectors <i>kernel trace · task ports</i> fs_usage · sample footprint · leaks

XNU — sysctl reads kernel variables · fs_usage needs root for the kernel tracing facility · a debugger needs the target's **task port**, given only if the target carries get-task-allow

What changed at the command line



defaults — preferences through cfprefsd

- Domain** = bundle id; -g = NSGlobalDomain (every app); -currentHost = this host only. A file *path* as domain: slated to go.
- No type flag ⇒ **string**: -bool -int -float -date -data -array(-add) -dict(-add); check with read-type. find <word> · export <dom> - (XML) · delete-all (every container).
- Since 10.8 cfprefsd caches and writes *asynchronously*: a hand-edited plist “will not necessarily be read, and may even be overwritten”.

UserDefaults lookup order — first hit wins (*volatile* = gone when the app quits):



launchctl — verbs on domain targets (U=\$<id> -u)

want	modern	legacy
load	bootstrap gui/\$U ~/Library/LaunchAgents/x.plist	load
unload	bootout gui/\$U/com.example.x	unload
restart	kickstart -k -p gui/\$U/com.example.x (-p: pid)	—
inspect	print gui/\$U/... blame (why it was launched)	list
keep off	disable gui/\$U/... — persists across boots	-w

- system/ (anyone reads, root modifies) · user/<uid> (exists without a login) · gui/<uid> (the login session: agents) · login/<asid> · pid/<pid> (the XPC services it can reach). Legacy verbs pick the domain by whether you are root.
- Relative ProgramArguments[0] resolves via _PATH_STDPATH = /usr/bin:/bin:/usr/sbin:/sbin, not your PATH; never daemon(3) / fork + exit.
- Background items (13+): sftool dumptbm lists, resetbtm resets. brew services = ~/Library/LaunchAgents (sudo: /Library/LaunchDaemons).

log — the unified log (10.12+)

- log stream --level debug = live (debug ⊃ info ⊃ default) · log show --last 1h = stored, default level unless --info --debug --last boot --style compact|json|ndjson.
- Keys: subsystem, category (with subsystem), process, sender (the library), messageType (default info debug error fault), eventMessage CONTAINS[c] “x”. ~/, logrc: default options + predicate aliases.
- Persist debug for one subsystem: sudo log config --subsystem com.example.app --mode “level:debug,persist:debug” (--status, --reset).
- sudo log collect --last 30m --output x.logarchive (--device: a paired device); store = /var/db/diagnostics. 27.0 archives need 26.2+ to read.
- <private> lifts only via OSLogPreferences (Info.plist) or a com.apple.system.logging.profile (Enable-Private-Data) — on a throwaway Mac.

Spotlight · open · clipboard

- mdfind 'kMDItemCFBundleIdentifier == "com.example.app"' — every copy (stale builds); onlyin <id> -name -live -0, mdls -name kMDItemWhereFroms f = download URL. sudo mdutil -E / rebuilds, -s status.
- open -a Xcode . -b <id> -n new instance · -W wait · -R reveal · -j hidden · -f no restore --env K=V --args ...
- pbpaste | jq . | pbcopy; pboard find. Without a UTF-8 LANG the C encoding is used — non-ASCII mangles.

Power — assertions held by powerd

caffeinate	assertion (pmset -g assertions)	note
-i	PreventUserIdleSystemSleep	default, no flag display
-d	PreventUserIdleDisplaySleep	
-s	PreventSystemSleep	AC power only
-m / -u	disk idle / “user is active”	-u: 5 s unless -t

caffeinate -i make holds it while make runs, -w <pid> until that pid exits. pmset -g log (sleep/wake history) · -g therm · -g sched; settings per source: -b battery -c charger -u UPS -a all; pmset sleepnow.

Storage — APFS

- A **container** shares free space among its **volumes**. macOS = a **volume group**: System (read-only 10.15, signed + sealed 11) + Data, joined by **firmlinks**.
- Snapshot = read-only copy of a volume. tmutil localsnapshot · listlocalsnapshots / · deletelocalsnapshots <date> · thinlocalsnapshots /; diskutil apfs listSnapshots /.
- cp -c = clonefile(2), copy-on-write. diskutil image create (26) writes **ASIF**, which replaces .sparseimage.

Network, DNS, users

scutil --dns --proxy scutil --get LocalHostName /etc/resolver/<domain>	resolver + proxy config (configd) also ComputerName, HostName per-domain DNS; resolv.conf is generated per service; empty clears service name ↔ en0 system lookup; -flushcache local directory node group membership
networksetup -setdnsservers Wi-Fi 9.9.9.9 networksetup -listallhardwareports dscacheutil -q host -a name example.com dscl . -list /Users UniqueID dseditgroup -o checkmember -m x admin	

Gatekeeper, quarantine, SIP, TCC

- spctl -a -vv -t execute App.app (install a .pkg, open a document): exit 0 ok · 1 failed · 2 bad args · **3 denied**. --global-disable only reveals “Anywhere” in Privacy & Security.
- Quarantine** = xattr com.apple.quarantine (not API); Safari sets it, Archive Utility propagates it; curl/scp don't set it, tar/unzip don't propagate. xattr -d com.apple.quarantine f · -r · -c (all).
- SIP guards /System /usr /bin /sbin /var — not /usr/local, /Applications, /Library. csrutil status (not an API); disable only in Recovery.
- tcutil reset AddressBook [bundle-id] re-prompts; 27: tcutil list (managed Macs); apps can no longer read the TCC database.

System info, Rosetta, sudo, Homebrew

- sw_vers --productVersion --buildVersion --productVersionExtra (Rapid Security Response). sysctl hw.nperfllevels, hw.perfllevel0, physicalcpu (0 = fastest cores), hw.memsize; sysctl.proc_translated 1 = under Rosetta.
- softwareupdate --install-rosetta: Rosetta stays general-purpose through 27, is not restored by the upgrade; sudo nvram boot-args="nox86exec=1" (26) makes x86 code crash — proof you no longer need it.
- Touch ID sudo: /etc/pam.d/sudo_local (14+, from sudo_local.template) survives updates.
- Homebrew: /opt/homebrew, no sudo after install; macOS 15+, Intel = Tier 3 (Oct 2026). brew bundle dump → Brewfile.

Process inspectors

sample <pid> — all threads' stacks, 10 s at 1 ms · spindump — the report of a force-quit hang · sudo fs_usage -w -f filesystems <proc> · footprint — *dirty* memory, phys_footprint = kernel ledger · vmmap --summary · heap · leaks --atExit -- ./tool: exit 0 clean, 1 leaks (a CI gate); reads Xcode .memgraph files too.

Traps

- ! bootstrap prints each failing path with its error; launchctl error 5 decodes the number. On 27 a downloaded (quarantined) plist won't load: xattr -d it.
- ! networksetup wants the service (“Wi-Fi”), not the device (en0).
- ! Leaks/vmmap on a distribution build: no task port (*unverified*) — Xcode adds get-task-allow only when building for debugging.

Related: xnu-launchd-xpc (plist keys, XPC) · macos-security-architecture (SIP, Gatekeeper, TCC) · logging-signposts-power (Logger) · macos-keychain-security-cli · notarization-distribution · systemd-deep (the Linux counterpart) · os-fundamentals